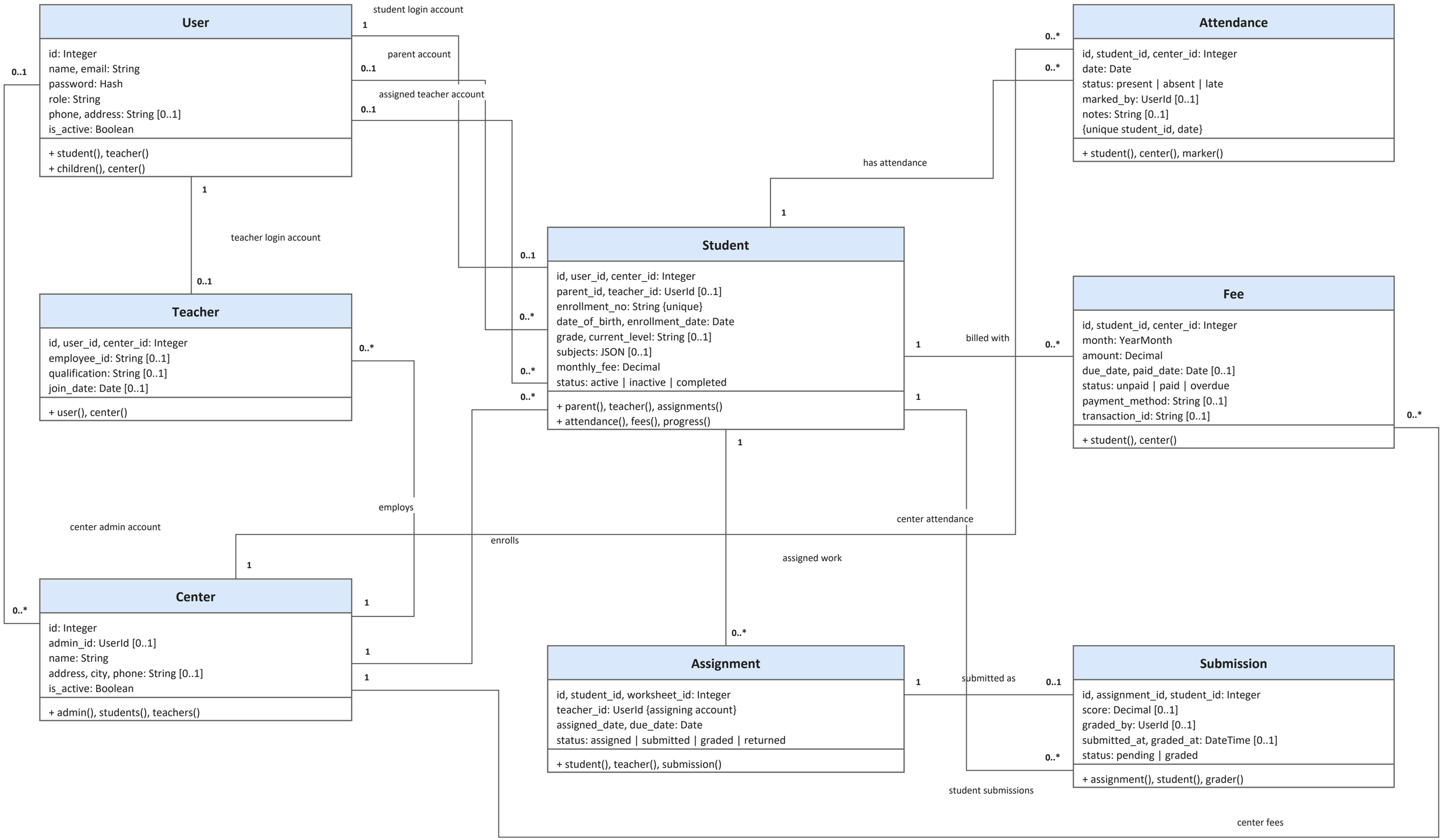
UML Class Diagram



Class diagram explanation

The diagram shows eight classes and their associations in the Center Admin module. The tables describe the classes and the multiplicities drawn between them.

Class	Role in the system	Notes and frontend evidence
User	Account identity and contact information for users of the system.	Stores name, email, password, role and active status. Relates to student and teacher profiles, children and administered centers.
Teacher	Teacher profile associated with a login account and a center.	Stores employee ID, qualification and join date. Its user() and center() relationships connect the profile to User and Center.
Center	Center record administered through a User account.	Stores name, address, city, phone and active status. Relates to its admin, students and teachers; attendance and fees also refer to the center.
Student	Student enrollment and placement record.	Stores enrollment number, dates, grade, level, subjects, monthly fee and status. Links to its login account, parent, teacher, center, attendance, fees, assignments and submissions.
Attendance	One student's attendance status on a date.	Stores Present, Absent or Late status, notes and marking user. Relates to Student and Center; the student/date pair is unique.
Fee	Monthly charge and payment details for a student.	Stores month, amount, due/paid dates, payment method and transaction ID. Status is Unpaid, Paid or Overdue. Relates to Student and Center.
Assignment	Work assigned to a student.	Stores worksheet and assigning-account IDs, assigned/due dates and work status. Relates to Student and an optional Submission.
Submission	Submitted work and its grading information.	Stores score, grading user, submitted/graded times and Pending or Graded status. Relates to Assignment and Student.

Class diagram explanation (continued)

Table. Relationships and multiplicities

Relationship	Multiplicity	Meaning
User - Student (student login account)	1 to 0..1	Each student has one login account; a user has at most one student profile in this diagram.
User - Teacher (teacher login account)	1 to 0..1	Each teacher profile has one login account; a user has at most one teacher profile in this diagram.
User - Student (parent account)	0..1 to 0..*	A student can have one linked parent account; a parent account can be linked to several students.
User - Student (assigned teacher account)	0..1 to 0..*	A student can have one assigned teacher account; the account can be assigned several students.
User - Center (center admin account)	0..1 to 0..*	A center can have one administrator account; the diagram allows a user to administer several centers.
Center - Teacher (employs)	1 to 0..*	A center can have many teacher profiles; each teacher profile belongs to one center.
Center - Student (enrolls)	1 to 0..*	A center can enroll many students; each student belongs to one center.
Student - Attendance (has attendance)	1 to 0..*	A student can accumulate many attendance records; each record belongs to one student.
Student - Fee (billed with)	1 to 0..*	A student can have many fee records; each fee belongs to one student.
Student - Assignment (assigned work)	1 to 0..*	A student can have many assignments; each assignment is for one student.
Assignment - Submission (submitted as)	1 to 0..1	An assignment can have one submission; each submission refers to one assignment.
Student - Submission (student submissions)	1 to 0..*	A student can have many submissions; each submission belongs to one student.
Center - Attendance (center attendance)	1 to 0..*	A center can have many attendance records; each record refers to one center.
Center - Fee (center fees)	1 to 0..*	A center can have many fee records; each fee refers to one center.